1

$a = 1$, $b = 2$, $c = 3$, found by inspection.

2

This is a linear equation, so $x = 4$ divided by 2, which is $x = 2$.

3(a)

The polynomial can be sketched as an upward curve crossing the x-axis around $x = 0$ and $x = 5$.

(b)

$5x - 2 = 4x + 1$, so $x = 3$ is the only solution.

4(a)

The arc length is proportional to the radius, so r is about 12 and θ is a right angle.

(b)

The shaded region is a semicircle, so area $= \frac{1}{2} \times \pi \times r^2 \approx 145$.